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Started	Tuesday, 13 October 2026, 4:22 PM
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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort cannot handle duplicate elements
- b. Bubble Sort is difficult to implement
- c. Bubble Sort does not guarantee sorted order
- d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **2**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of duplicate keys
- b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of all keys
- d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Merge sort
- b. Heap sort
- c. Quick sort
- d. Selection sort

The correct answer is:
Merge sort

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is sorted independently
- b. Each part is searched for a specific element
- c. The parts are merged without sorting
- d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting is rarely used in programming
- c. Sorting decreases the efficiency of selection operations
- d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Quick Sort
- c. Merge Sort
- d. Insertion Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It avoids the need for recursion
- b. It only works on small datasets
- c. It allows for efficient parallel processing and sorting of data
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **9**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It increases the number of comparisons needed
- b. It allows for faster searching
- c. It makes the search operation slower
- d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **10**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Bubble Sort
- b. Binary Search
- c. Linear Search
- d. Merge Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It complicates the selection of items
- b. It makes it easier to select items based on their relationship to the rest of the items
- c. It makes the data unsorted
- d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Combining sorted sublists
- c. Sorting the entire list sequentially
- d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Binary Search
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It repeatedly finds the minimum element and moves it to the sorted part of the list
- b. It compares adjacent elements and swaps them if necessary
- c. It builds a sorted array one element at a time
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **15**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. order()
- c. sorted()
- d. arrange()

The correct answer is: sort()